

COAL LAB TASK # 6

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SAP ID # 70150006 SECTION C

TASK 1:

```
org 100h
```

```
.data
```

```
str1 db 0ah,0dh,"Enter a number: $"
```

```
gt db 0ah,0dh,"Num 1 is greater than Num 2 $"
```

```
lt db 0ah,0dh,"Num 1 is less than Num 2 $"
```

```
eq db 0ah,0dh,"Num 1 is equal to Num 2 $"
```

```
.code
```

```
mov ah,09h
```

```
lea dx,str1
```

```
int 21h
```

```
mov ah,01h
```

```
int 21h
```

```
mov bl,al
```

```
mov ah,09h
```

```
lea dx,str1
```

```
int 21h
```

mov ah,01h

int 21h

cmp bl,al

je equal

jg greater

jl less

equal:

mov ah,09h

lea dx,eq

int 21h

jmp ex

greater:

mov ah,09h

lea dx,gt

int 21h

jmp ex

less:

mov ah,09h

lea dx,lt

int 21h

jmp ex

ex:

mov ah,4ch

int 21h

ret

TASK#2

.MODEL SMALL

.STACK 100H

.DATA

VAR DB 20 DUP('\$')

COUNT DB 0

.CODE

MAIN PROC

MOV AX,@DATA

MOV DS,AX

MOV AL,11010101B

ROR AL,1

XOR CL,CL

MOV CH,8

CHECK:

```
ROR AL,1
JNC SKIP
INC CL

SKIP:
DEC CH
JNZ CHECK
;ADD CL,20H
;MOV COUNT,CL

MOV AH,02H
LEA DX,COUNT
INT 21H

MOV AH,4CH
INT 21H

MAIN ENDP
END MAIN
```

TASK#3

```
.MODEL SMALL
.STACK 100H
.DATA
VAR DB 20 DUP('$')
COUNT DB 0
```

.CODE

MAIN PROC

MOV AX,@DATA

MOV DS,AX

MOV AL,11010101B

ROR AL,1

XOR CL,CL

MOV CH,8

CHECK:

ROR AL,1

JNC SKIP

INC CL

SKIP:

DEC CH

JNZ CHECK

;ADD CL,20H

;MOV COUNT,CL

MOV AH,02H

LEA DX,COUNT

INT 21H

MOV AH,4CH

INT 21H

MAIN ENDP

END MAIN